Beta Diversity Analysis

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Contents

Beta diversity differences by age, sex, and cohort

```r
library(lme4)
```

```
## Loading required package: Matrix
```

```r
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
library(knitr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(phyloseq)
library(vegan)
```

```
## Loading required package: permute
```

```r
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(tibble)
library(reshape2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.2     v stringr   1.5.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(qiime2R)
library(ggpubr)
```

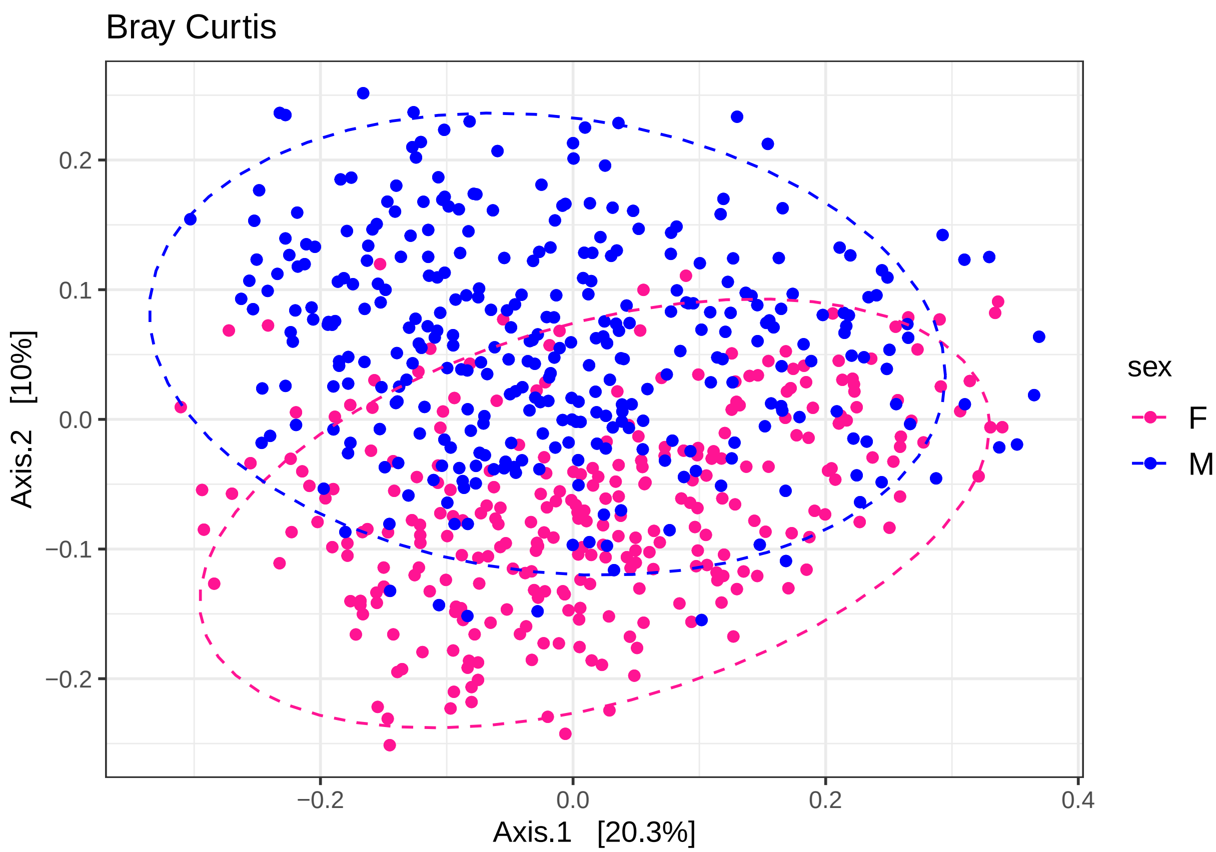
```
##
## Attaching package: 'ggpubr'
##
## The following object is masked from 'package:qiime2R':
##
##     mean_sd
```

```r
library(ggplot2)
library(pairwiseAdonis)
```

```
## Loading required package: cluster
```

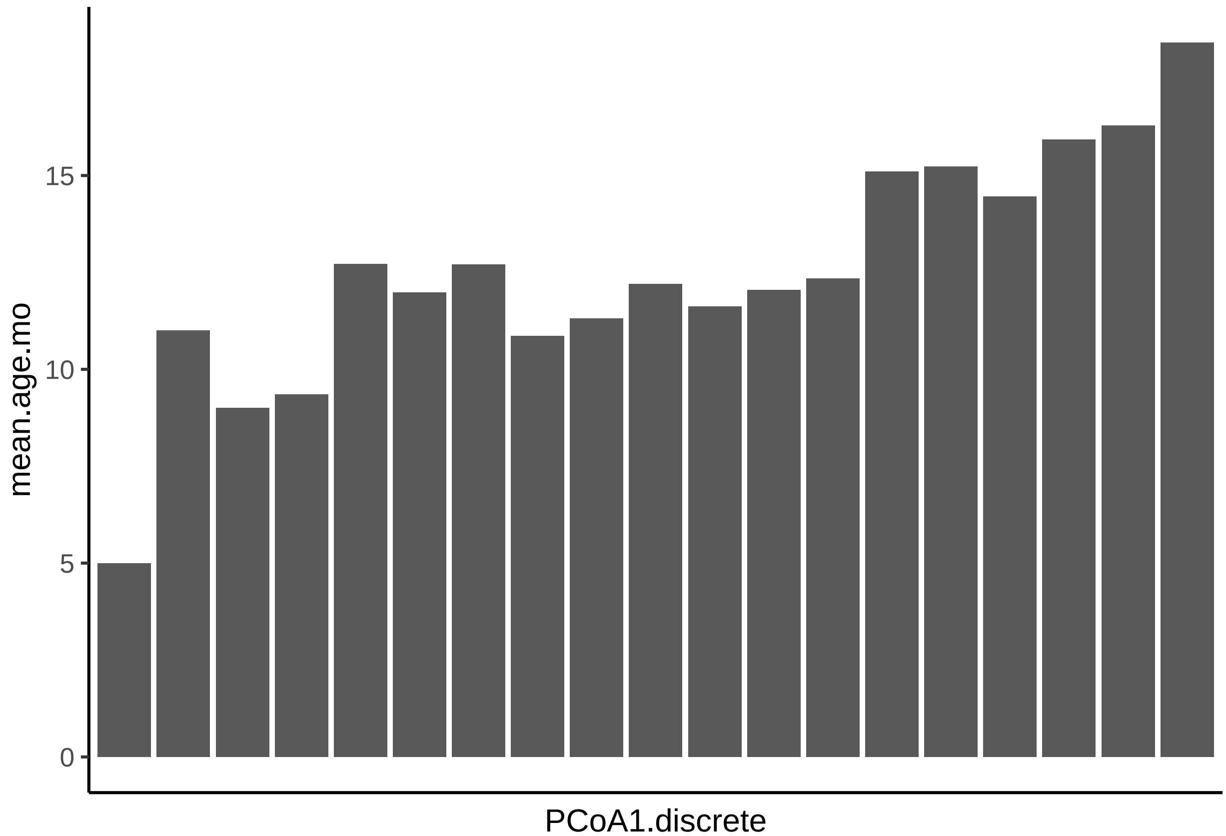
```r
theme_set(theme_bw())
load("qiime_f.RData")
qiime_norm<-  qiime_f %>%
  transform_sample_counts(function(x) x/sum(x))
```

Bray-Curtis distance PCoA

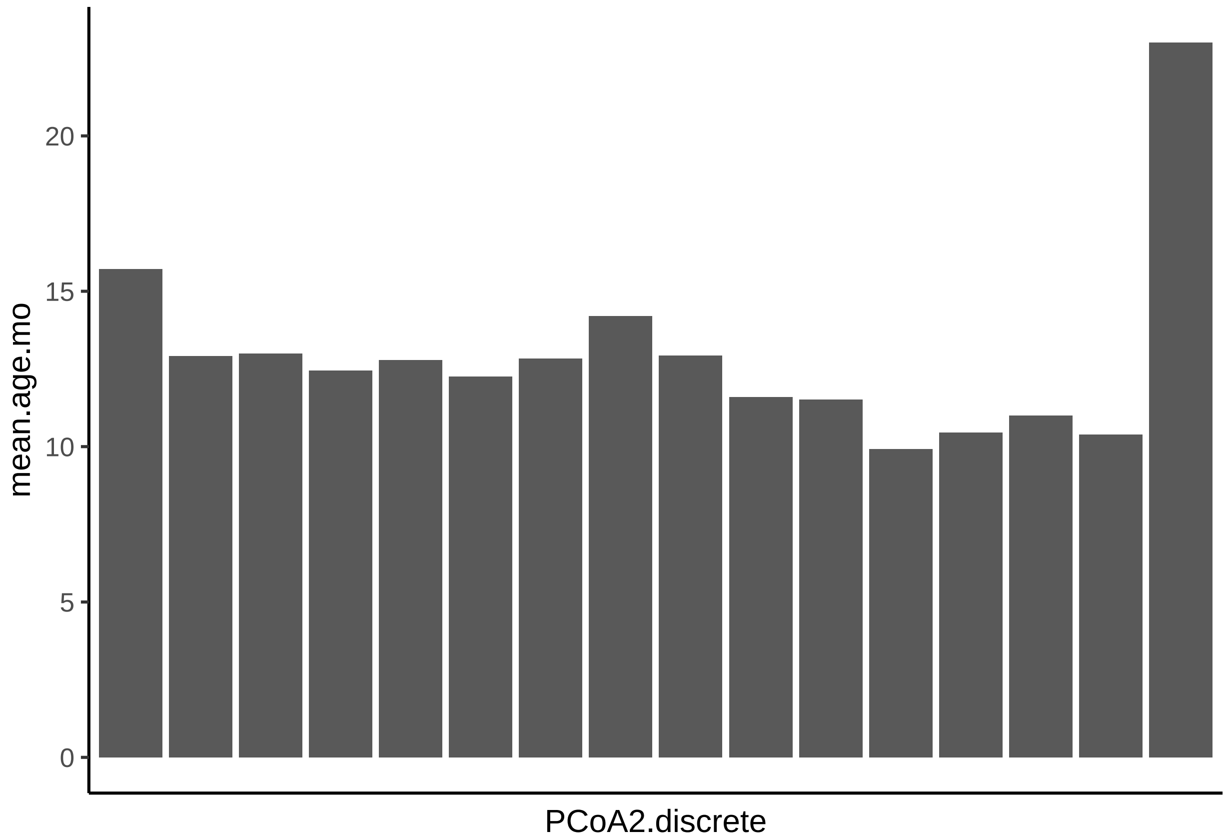



```
samples_df<-as.data.frame(sample_data(qiime_f))
bray_dist_matrix<- phyloseq::distance(qiime_norm, method ="bray")
pcoa.bc <- cmdscale(bray_dist_matrix, eig = TRUE, k = 3)

pcoa.bc.df <- merge(
  data.frame(pcoa.bc$points) %>% setNames(c("PCoA1", "PCoA2", "PCoA3")),
  samples_df,
  by="row.names") %>%
  mutate(sex=factor(sex, levels=c("M","F")),
         sex=factor(sex, levels=c("M","F")))
frac.var.explained.by.pcos <- pcoa.bc$eig / sum(pcoa.bc$eig)

full.pcoa.p <- pcoa.bc.df %>%
    arrange(desc(month_n)) %>%
  ggplot(aes(x=PCoA1, y=PCoA2, color=sex)) +
  geom_point(shape=16, alpha=0.8) +
  scale_color_manual(values=c("#3639ff","deeppink")) +
  scale_size_continuous(breaks=c(5,10,16,22,28,34,40,46), range=c(0.25,2))+
  stat_ellipse(linetype="dashed")

#Extract the x-axis range from the PCoA plot
pcoa1.lims <- layer_scales(full.pcoa.p, 1, 1)$x$range$range

#Create break points for binning, whener length.out=20 reates 20 equally spaced intervals across the PC
```

```r
pcoa1.intervals <- seq(from=pcoa1.lims[1], to=pcoa1.lims[2], length.out=20)

pcoa1.age.barplots <- pcoa.bc.df %>%
  mutate(PCoA1.discrete=cut(PCoA1, pcoa1.intervals, include.lowest=T)) %>%
  group_by(PCoA1.discrete) %>%
  summarise(mean.age.mo=mean(month_n), .groups="drop") %>%
  ggplot(aes(x=PCoA1.discrete, y=mean.age.mo)) +
  geom_col() +
  theme_classic(base_size=12) +
  theme(axis.text.x=element_blank(),
        axis.ticks.x=element_blank())

#Extract the y-axis range from the PCoA plot
pcoa2.lims <- layer_scales(full.pcoa.p, 1, 1)$y$range$range
pcoa2.intervals <- seq(from=pcoa2.lims[1], to=pcoa1.lims[2], length.out=20)

pcoa2.age.barplots <- pcoa.bc.df %>%
  mutate(PCoA2.discrete=cut(PCoA2, pcoa2.intervals, include.lowest=T)) %>%
  group_by(PCoA2.discrete) %>%
  summarise(mean.age.mo=mean(month_n), .groups="drop") %>%
  ggplot(aes(x=PCoA2.discrete, y=mean.age.mo)) +
  geom_col() +
  theme_classic(base_size=12) +
  theme(axis.text.x=element_blank(),
        axis.ticks.x=element_blank())

pcoa1.age.barplots
```



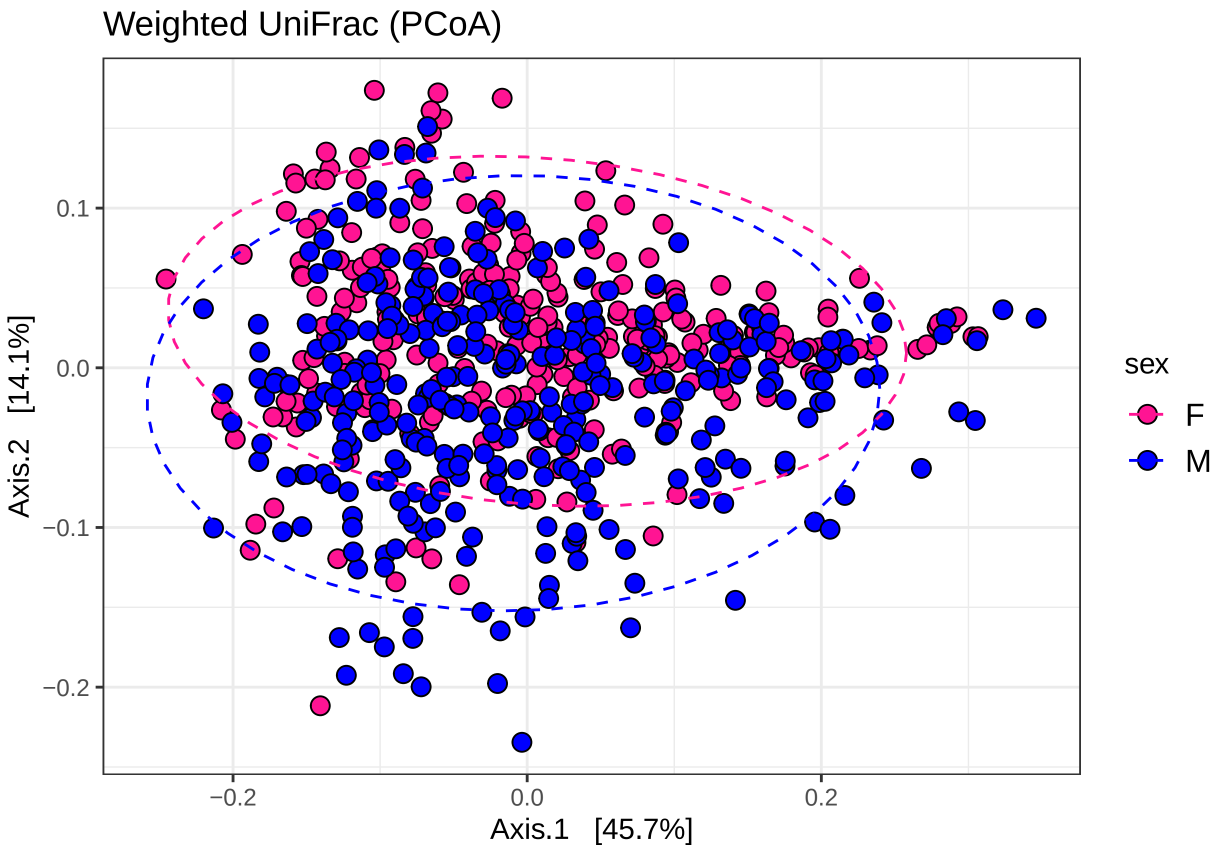

```
pcoa2.age.barplots
```

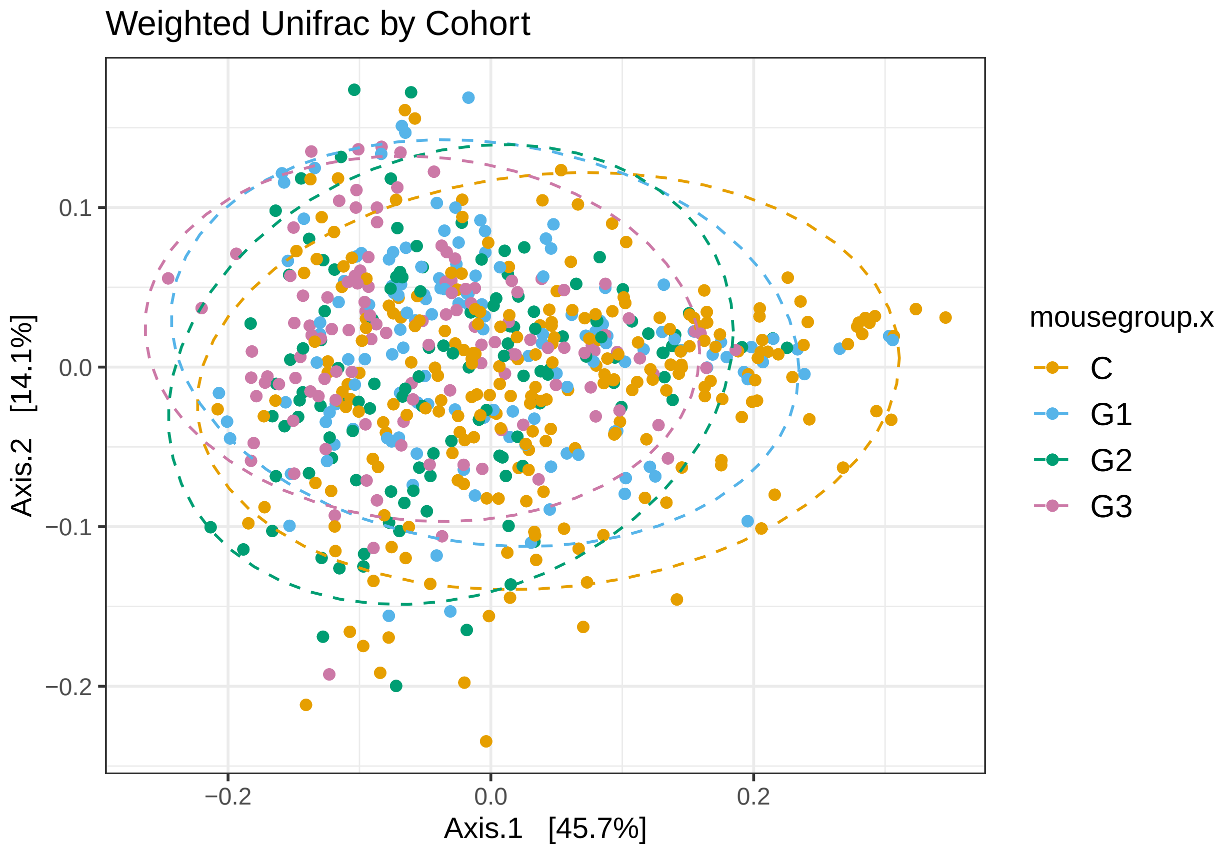



```r
ordination_wu <- ordinate(qiime_norm, method = "PCoA", distance = "unifrac", weighted=TRUE)
```

```
## Warning in matrix(tree$edge[order(tree$edge[, 1]), ][, 2], byrow = TRUE, : data
## length [8563] is not a sub-multiple or multiple of the number of rows [4282]
```

Weighted UniFrac PCoA

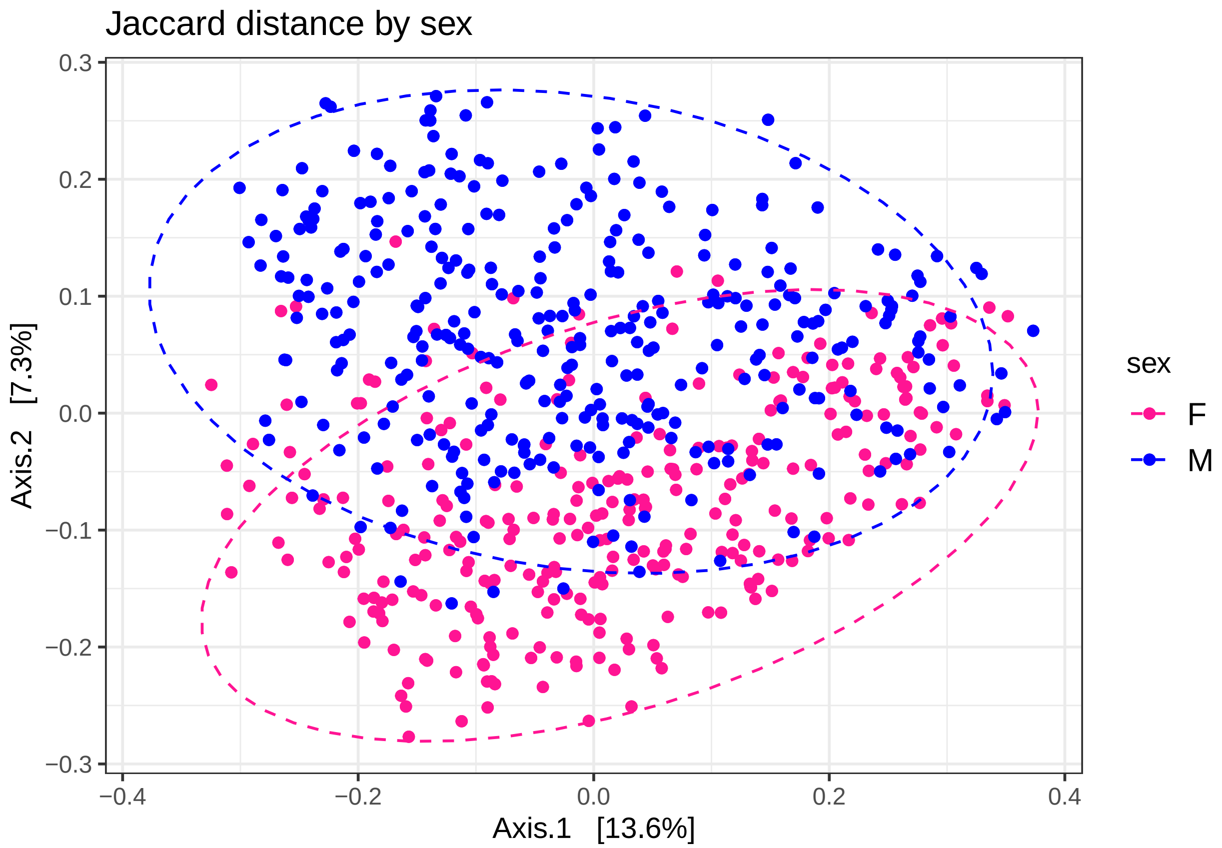




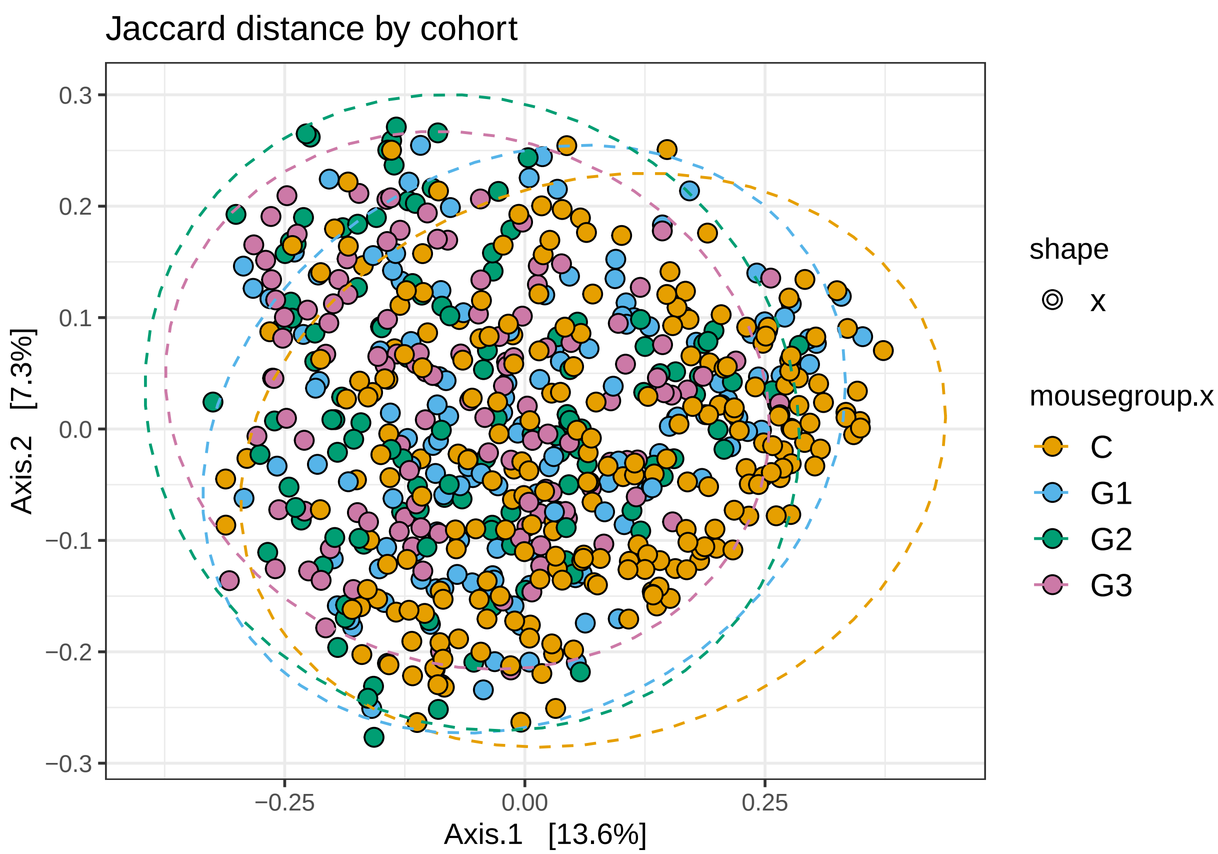
Jacard Distance

```
ordination_jac <- ordinate(qiime_norm, method = "PCoA", distance = "jaccard")
```

```
plot_ordination(qiime_norm, ordination_jac, color="sex", title="Jaccard distance by sex")+geom_point(aes
  scale_fill_manual(values=c("deeppink","blue"))+theme(legend.text = element_text(size=12))+
 scale_color_manual(values=c("deeppink","blue"))+
  guides(
    fill = guide_legend(override.aes = list(shape = 21)),
    shape = guide_legend())+
  stat_ellipse(aes(group=sex), linetype="dashed")
```



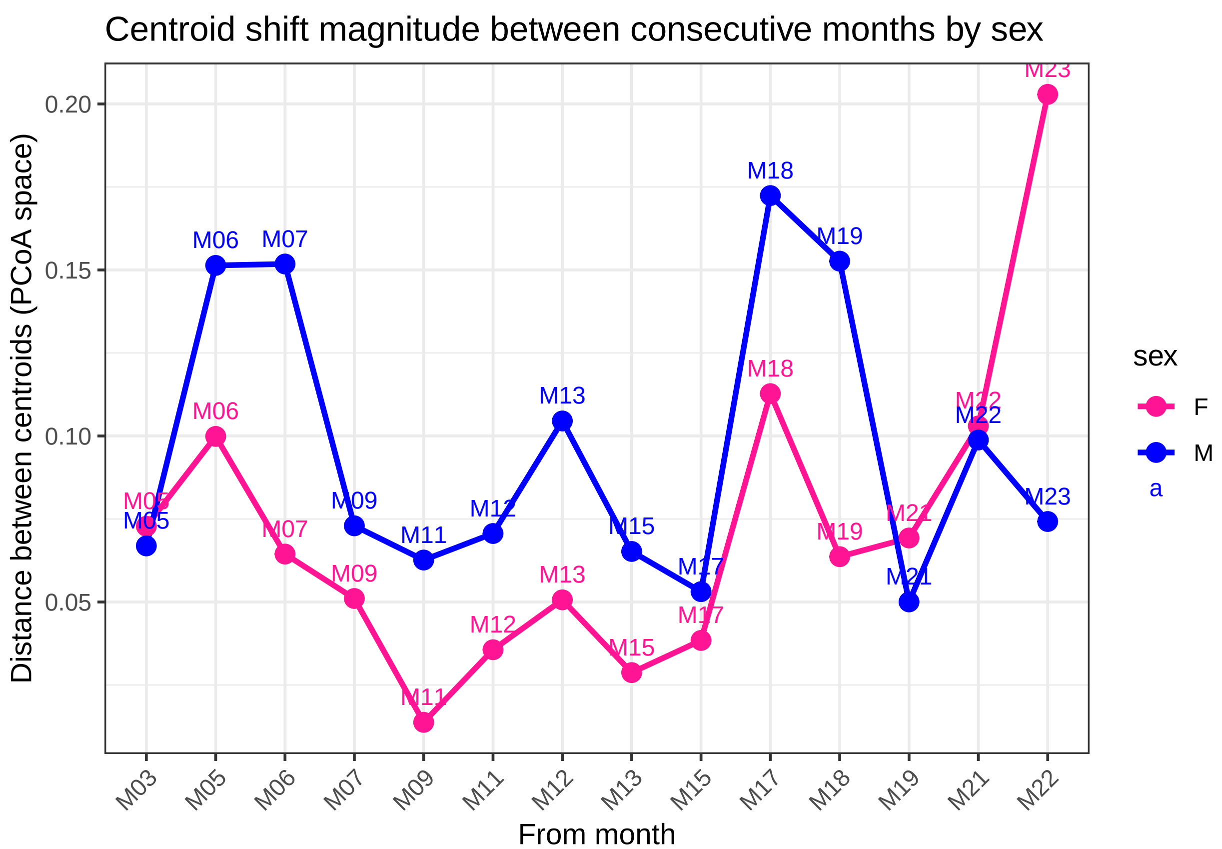

```
plot_ordination(qiime_norm, ordination_jac, color="mousegroup.x", title="Jaccard distance by cohort")+g
  scale_fill_manual(values=c("#E69F00", "#56B4E9", "#009E73", "#CC79A7"))+theme(legend.text = element_t
  scale_shape_manual(values = c(21, 22, 23, 24, 25))+scale_color_manual(values=c("#E69F00", "#56B4E9",
  guides(
    fill = guide_legend(override.aes = list(shape = 21)),
    shape = guide_legend()
  )+stat_ellipse(aes(group=mousegroup.x), linetype="dashed")
```



How does the beta diversity distance change with time in males and females

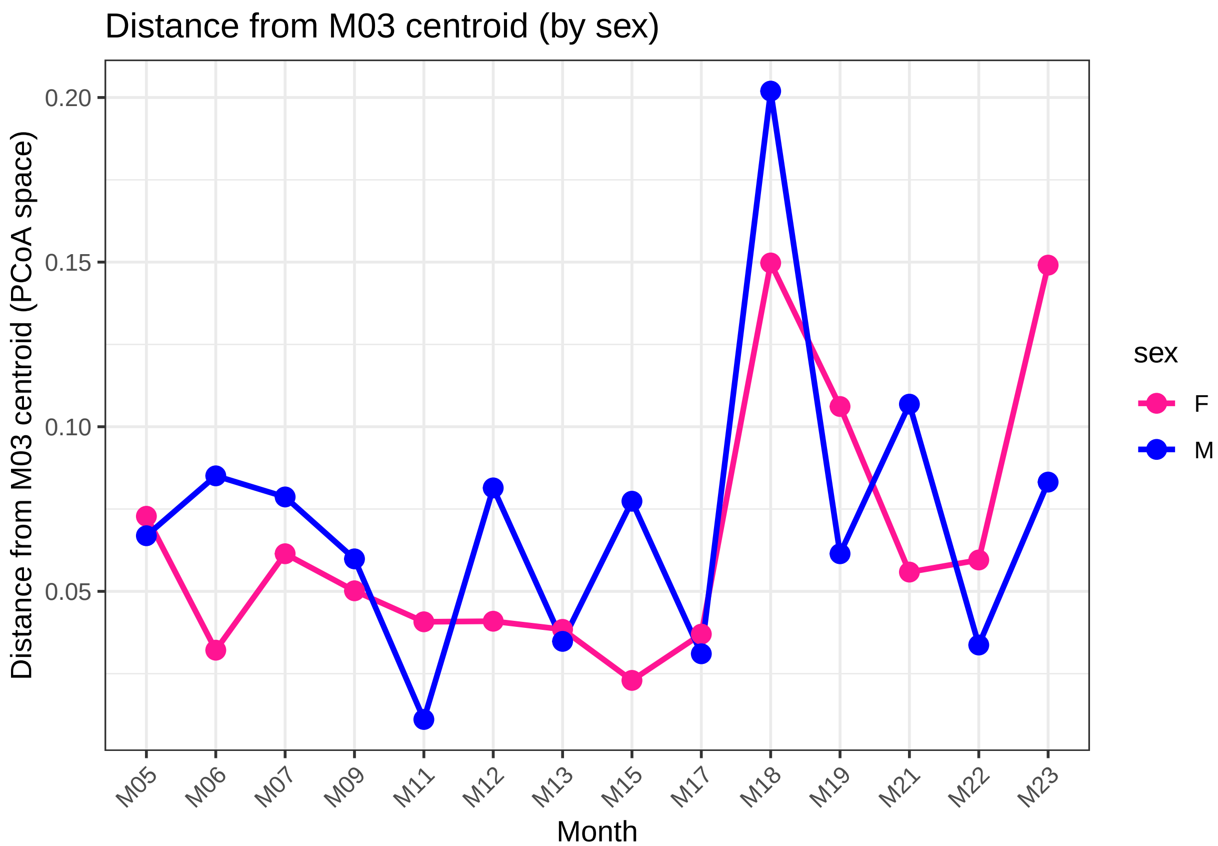
```
ord_df <- as.data.frame(bdist.ord$vectors)

samples_df<-as.data.frame(sample_data(qiime_f))

ord_df$month <- samples_df$month
ord_df$sex<-samples_df$sex

#Calculating the average distance of centroid for each month and sex
centroids <- ord_df %>%
  group_by(month, sex) %>%
  summarise(
    PC1 = mean(Axis.1),
    PC2 = mean(Axis.2),
    .groups = "drop"
  ) %>%
  arrange(sex, as.numeric(gsub("M", "", month))) %>%
  mutate(month = factor(month, levels = unique(month)))

#How do the centroids change with time
centroid_shift <- centroids %>%
  group_by(sex) %>%
  arrange(sex, as.numeric(gsub("M", "", month))) %>%
```

```
  mutate(
    next_month = lead(month),
    next_PC1 = lead(PC1),
    next_PC2 = lead(PC2),
    shift_distance = sqrt((next_PC1 - PC1)^2 + (next_PC2 - PC2)^2)
  ) %>%
  filter(!is.na(next_month)) %>%
  ungroup()
```




```
# Centroid to M03====
centroid_distance_from_M03 <- centroids %>%
  group_by(sex) %>%
  mutate(
    M03_PC1 = PC1[month == "M03"],
    M03_PC2 = PC2[month == "M03"],
    dist_from_M03 = sqrt((PC1 - M03_PC1)^2 + (PC2 - M03_PC2)^2)
  ) %>%
  ungroup() %>%
  filter(month != "M03")


ggplot(centroid_distance_from_M03,
       aes(x = month, y = dist_from_M03, color = sex, group = sex)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
```

```
  theme_bw() +
  labs(
    title = "Distance from M03 centroid (by sex)",
    x = "Month",
    y = "Distance from M03 centroid (PCoA space)"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))+
  scale_color_manual(values=c("deeppink","blue"))
```



As mice ages, within sample distances increases with time

```
bray_dist<-phyloseq::distance(qiime_norm, method = "bray")
samples_df$SampleID<-rownames(samples_df)
samples_df$sex_age <-paste(samples_df$sex,samples_df$month,sep="_")
bd <- betadisper(bray_dist, samples_df$sex_age)
bd_df <- data.frame(
  Distance = bd$distances,
  Month = samples_df$month,
  Sex = samples_df$sex,
  #cohort=sampledf$mousegroup.x,
  mouse= samples_df$mouse
)
```

```r
#Optional to make a line plot with medians
median_df <- bd_df %>%
  group_by(Sex, Month) %>%
  summarise(median_distance = median(Distance, na.rm = TRUE), .groups = "drop")

anova(bd)
```

```
## Analysis of Variance Table
##
## Response: Distances
##            Df Sum Sq   Mean Sq F value    Pr(>F)
## Groups     29 0.4518 0.0155792  4.9745 4.28e-15 ***
## Residuals 594 1.8603 0.0031318
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

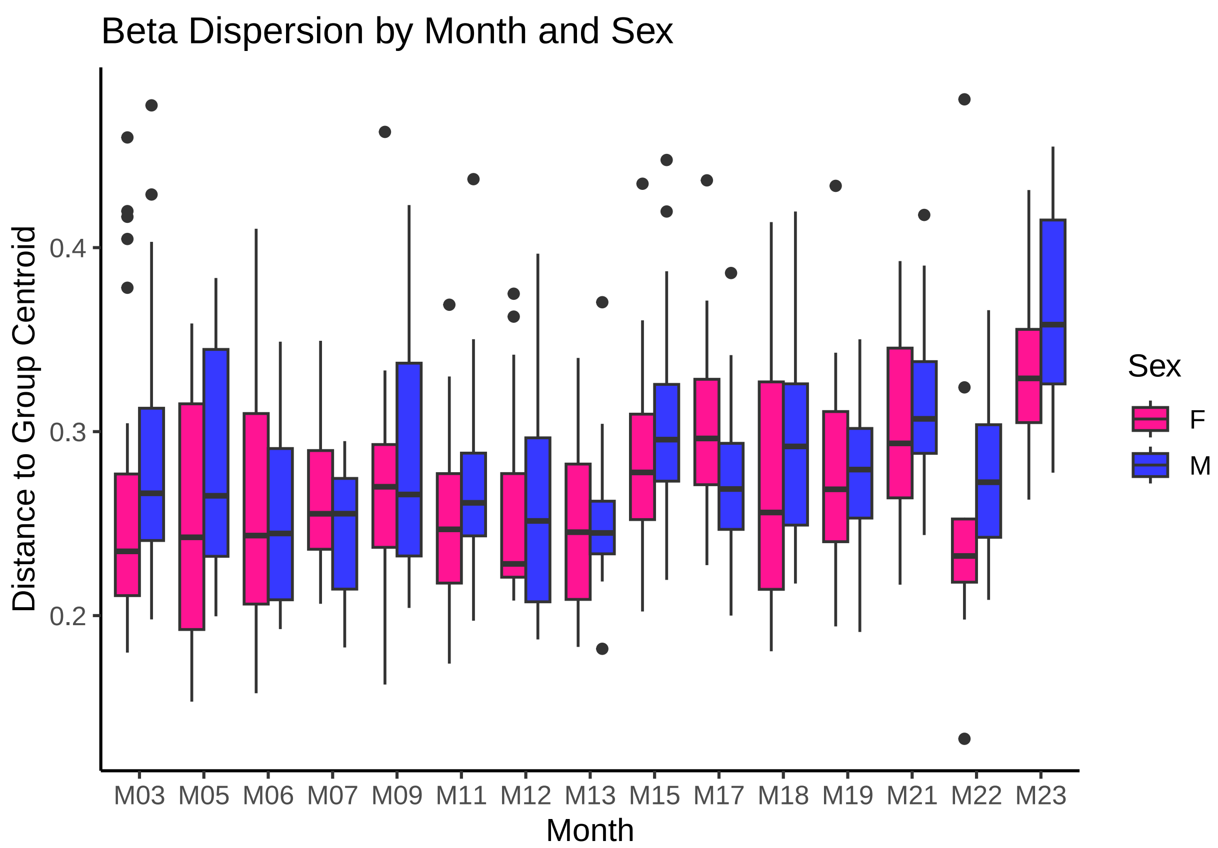
```r
TukeyHSD(aov(bd$distances ~ samples_df$month))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = bd$distances ~ samples_df$month)
##
## $`samples_df$month`
##                  diff           lwr        upr     p adj
## M05-M03 -0.0086178653 -0.0464273772 0.02919165 0.9999822
## M06-M03 -0.0211132771 -0.0620066596 0.01978011 0.9135083
## M07-M03 -0.0224056768 -0.0587744031 0.01396305 0.7360163
## M09-M03  0.0002156014 -0.0317319553 0.03216316 1.0000000
## M11-M03 -0.0148745290 -0.0517849701 0.02203591 0.9893875
## M12-M03 -0.0159848057 -0.0578640291 0.02589442 0.9937477
## M13-M03 -0.0252390521 -0.0644442156 0.01396611 0.6703915
## M15-M03  0.0170482038 -0.0151467436 0.04924315 0.8963544
## M17-M03  0.0138237656 -0.0261832225 0.05383075 0.9977522
## M18-M03  0.0115191806 -0.0333959768 0.05643434 0.9999245
## M19-M03  0.0005587248 -0.0390375874 0.04015504 1.0000000
## M21-M03  0.0319818672 -0.0068502606 0.07081400 0.2470436
## M22-M03 -0.0082437227 -0.0555269289 0.03903948 0.9999994
## M23-M03  0.0744139220  0.0383007767 0.11052707 0.0000000
## M06-M05 -0.0124954118 -0.0585791235 0.03358830 0.9998532
## M07-M05 -0.0137878115 -0.0559081629 0.02833254 0.9987440
## M09-M05  0.0088334667 -0.0295343079 0.04720124 0.9999798
## M11-M05 -0.0062566637 -0.0488456342 0.03633231 0.9999999
## M12-M05 -0.0073669404 -0.0543276591 0.03959378 0.9999999
## M13-M05 -0.0166211868 -0.0612136143 0.02797124 0.9950717
## M15-M05  0.0256660691 -0.0129079427 0.06424008 0.6164597
## M17-M05  0.0224416309 -0.0228573629 0.06774062 0.9365383
## M18-M05  0.0201370459 -0.0295500863 0.06982418 0.9888003
## M19-M05  0.0091765901 -0.0357601174 0.05411330 0.9999955
## M21-M05  0.0405997325 -0.0036650830 0.08486455 0.1134731
## M22-M05  0.0003741426 -0.0514635014 0.05221179 1.0000000
## M23-M05  0.0830317873  0.0411319185 0.12493166 0.0000000
## M07-M06 -0.0012923997 -0.0462015695 0.04361677 1.0000000
```

```
## M09-M06  0.0213288785 -0.0200812128 0.06273897 0.9150076
## M11-M06  0.0062387481 -0.0391102315 0.05158773 1.0000000
## M12-M06  0.0051284715 -0.0443489844 0.05460593 1.0000000
## M13-M06 -0.0041257750 -0.0513612927 0.04310974 1.0000000
## M15-M06  0.0381614809 -0.0034397681 0.07976273 0.1133536
## M17-M06  0.0349370427 -0.0129660718 0.08284016 0.4538336
## M18-M06  0.0326324577 -0.0194398060 0.08470472 0.7114758
## M19-M06  0.0216720019 -0.0258886669 0.06923267 0.9677945
## M21-M06  0.0530951444  0.0061687825 0.10002151 0.0108646
## M22-M06  0.0128695545 -0.0412585425 0.06699765 0.9999700
## M23-M06  0.0955271991  0.0508247546 0.14022964 0.0000000
## M09-M07  0.0226212782 -0.0143274862 0.05957004 0.7446840
## M11-M07  0.0075311477 -0.0337840488 0.04884634 0.9999990
## M12-M07  0.0064208711 -0.0393877979 0.05222954 1.0000000
## M13-M07 -0.0028333753 -0.0462109001 0.04054415 1.0000000
## M15-M07  0.0394538806  0.0022910034 0.07661676 0.0252485
## M17-M07  0.0362294424 -0.0078741164 0.08033300 0.2509635
## M18-M07  0.0339248574 -0.0146748974 0.08252461 0.5329385
## M19-M07  0.0229644016 -0.0207669686 0.06669577 0.9022929
## M21-M07  0.0543875440  0.0113468777 0.09742821 0.0018288
## M22-M07  0.0141619541 -0.0366343686 0.06495828 0.9997963
## M23-M07  0.0968195988  0.0562151154 0.13742408 0.0000000
## M11-M09 -0.0150901304 -0.0525722274 0.02239197 0.9894861
## M12-M09 -0.0162004071 -0.0585843225 0.02618351 0.9936590
## M13-M09 -0.0254546535 -0.0651984804 0.01428917 0.6784540
## M15-M09  0.0168326024 -0.0160161658 0.04968137 0.9181489
## M17-M09  0.0136081642 -0.0269268334 0.05414316 0.9983483
## M18-M09  0.0113035792 -0.0340825232 0.05668968 0.9999470
## M19-M09  0.0003431234 -0.0397866021 0.04047285 1.0000000
## M21-M09  0.0317662658 -0.0076096288 0.07114216 0.2791150
## M22-M09 -0.0084593240 -0.0561901163 0.03927147 0.9999993
## M23-M09  0.0741983206  0.0375010973 0.11089554 0.0000000
## M12-M11 -0.0011102766 -0.0473502006 0.04512965 1.0000000
## M13-M11 -0.0103645231 -0.0541972284 0.03346818 0.9999719
## M15-M11  0.0319227329 -0.0057704476 0.06961591 0.2067231
## M17-M11  0.0286982947 -0.0158530275 0.07324962 0.6694253
## M18-M11  0.0263937097 -0.0226127447 0.07540016 0.8834533
## M19-M11  0.0154332539 -0.0287496518 0.05961616 0.9974791
## M21-M11  0.0468563963  0.0033570243 0.09035577 0.0210611
## M22-M11  0.0066308064 -0.0445547659 0.05781638 1.0000000
## M23-M11  0.0892884510  0.0481980571 0.13037885 0.0000000
## M13-M12 -0.0092542464 -0.0573457712 0.03883728 0.9999979
## M15-M12  0.0330330095 -0.0095376911 0.07560371 0.3437180
## M17-M12  0.0298085713 -0.0189388287 0.07855597 0.7463497
## M18-M12  0.0275039863 -0.0253460021 0.08035397 0.9083891
## M19-M12  0.0165435305 -0.0318673965 0.06495446 0.9980018
## M21-M12  0.0479666729  0.0001787659 0.09575458 0.0480519
## M22-M12  0.0077410830 -0.0471356106 0.06261778 1.0000000
## M23-M12  0.0903987277  0.0447927064 0.13600475 0.0000000
## M15-M13  0.0422872559  0.0023442961 0.08223022 0.0261247
## M17-M13  0.0390628177 -0.0074074362 0.08553307 0.2171663
## M18-M13  0.0367582327 -0.0139990009 0.08751547 0.4666170
## M19-M13  0.0257977769 -0.0203193927 0.07191495 0.8501476
## M21-M13  0.0572209193  0.0117581923 0.10268365 0.0019644
```

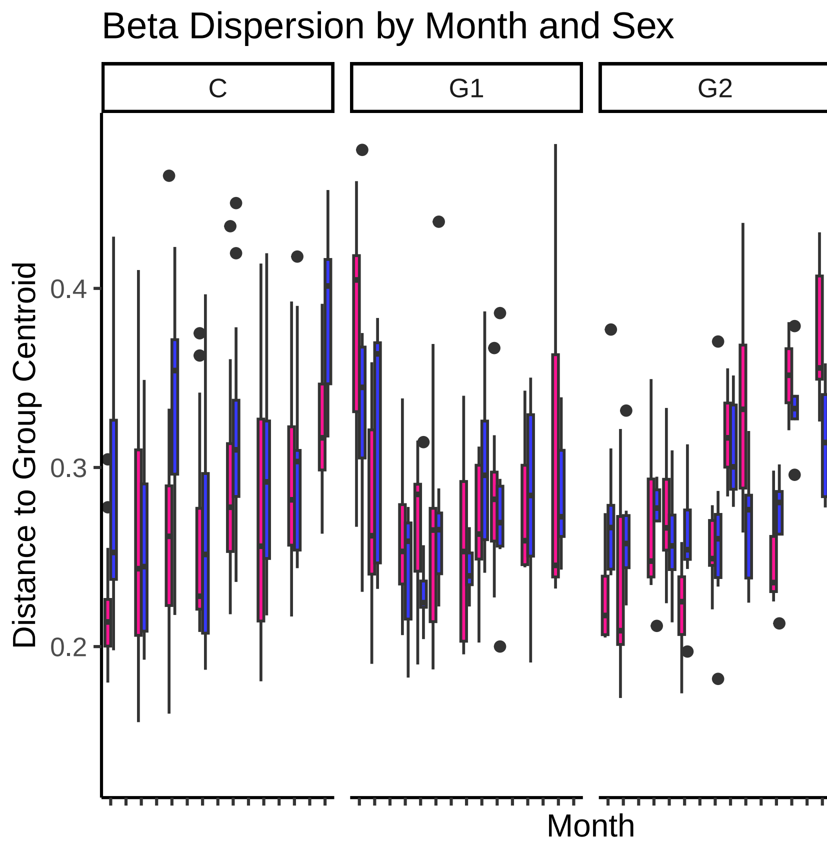
```
## M22-M13  0.0169953294 -0.0358689023 0.06985956 0.9989696
## M23-M13  0.0996529741  0.0564895097 0.14281644 0.0000000
## M17-M15 -0.0032244382 -0.0439547008 0.03750582 1.0000000
## M18-M15 -0.0055290232 -0.0510896043 0.04003156 1.0000000
## M19-M15 -0.0164894790 -0.0568164319 0.02383747 0.9878180
## M21-M15  0.0149336634 -0.0246432154 0.05451054 0.9944245
## M22-M15 -0.0252919265 -0.0731886569 0.02260480 0.8983732
## M23-M15  0.0573657182  0.0204529229 0.09427851 0.0000173
## M18-M17 -0.0023045850 -0.0536836762 0.04907451 1.0000000
## M19-M17 -0.0132650408 -0.0600657629 0.03353568 0.9997531
## M21-M17  0.0181581016 -0.0279978709 0.06431407 0.9915827
## M22-M17 -0.0220674883 -0.0755290749 0.03139410 0.9866893
## M23-M17  0.0605901564  0.0166971169 0.10448320 0.0003152
## M19-M18 -0.0109604558 -0.0620204191 0.04009951 0.9999916
## M21-M18  0.0204626866 -0.0300069683 0.07093234 0.9887560
## M22-M18 -0.0197629033 -0.0769901128 0.03746431 0.9977655
## M23-M18  0.0628947414  0.0144859490 0.11130353 0.0010887
## M21-M19  0.0314231424 -0.0143773231 0.07722361 0.5635868
## M22-M19 -0.0088024475 -0.0619574104 0.04435252 0.9999997
## M23-M19  0.0738551972  0.0303361467 0.11737425 0.0000012
## M22-M21 -0.0402255899 -0.0928137663 0.01236259 0.3686056
## M23-M21  0.0424320548 -0.0003928675 0.08525698 0.0550652
## M23-M22  0.0826576447  0.0320439962 0.13327129 0.0000041
```

```r
ggplot(bd_df, aes(x = Month, y = Distance, fill = Sex)) +
  geom_boxplot(position = position_dodge(width = 0.75)) +
  theme_classic(base_size = 12) +
  labs(title = "Beta Dispersion by Month and Sex",
       x = "Month", y = "Distance to Group Centroid") +
  scale_fill_manual(values = c("M" = "#3639ff", "F" = "deeppink")) +
  theme(legend.position = "right")
```




```
bd_df_cohort <- data.frame(
  Distance = bd$distances,
  Month = samples_df$month,
  Sex = samples_df$sex,
  cohort=samples_df$mousegroup.x,
  mouse= samples_df$mouse
)
```

```
ggplot(bd_df_cohort, aes(x = Month, y = Distance, fill = Sex)) +
  geom_boxplot(position = position_dodge(width = 0.75)) +
  theme_classic(base_size = 12) + facet_grid(~cohort)+
  labs(title = "Beta Dispersion by Month and Sex",
       x = "Month", y = "Distance to Group Centroid") +
  scale_fill_manual(values = c("M" = "#3639ff", "F" = "deeppink")) +
  theme(legend.position = "right")+theme(axis.text.x=element_blank())
```



This pattern was consistent with each cohort